Chi Zhang

System & Function Development Engineer Candidate

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Education

Shanghai Jiao Tong University — School of Aeronautics and Astronautics Sep 2023 – Mar 2026
M.Eng., Aerospace Systems Engineering (Vehicle System Design direction) *Shanghai, China*

- Thesis: *Research on a Vehicle Data Management System based on the MBSE ARCADIA Methodology* — see Project Experience.

Moscow Aviation Institute (MAI) Sep 2024 – Jul 2025
M.Eng., Aviation Engineering (English-taught program) *Moscow, Russia*

Northwestern Polytechnical University — School of Automation Sep 2019 – Jun 2023
B.Eng., Automation (Navigation, Guidance & Control direction) *Xi'an, China*

Work Experience (Internships)

Shanghai Civil Avionics System Co., Ltd. — Integration & Verification (I&V) Engineer Jul 2026 – Present
Integration & Verification Department *Shanghai, China*

- Researching the **ARINC 615A** (TFTP-based data-loading protocol) / **ARINC 665** (LSAP software-packaging) stack for the DCAS display system as an **OHMS** member system; authored an internal training deck (615A session model, Wireshark-based ARINC 664/TFTP analysis) and defined a verification strategy favoring requirement-traceability review and Review Cases over exhaustive test-case coverage for this point-to-point interaction.
- Supporting integration of a flight test article ("Red Label 3.0") by executing test procedures and tracking problem reports (PRs) to closure; using an OHMS avionics simulator to validate IDU data-loading/fault-history-retrieval functions and deriving system-level test cases toward a reusable ARINC 615A/665 conformance-test capability.

AVIC Airborne Systems Co., Ltd. — System Engineer Jul 2025 – Sep 2025

- Supported QA of onboard power-distribution-system documentation; conducted cross-review/consistency analysis across design docs, verification reports, and test cases for new projects.

Project Experience

UAV Flight Management System — Algorithm Portfolio (personal project, ongoing) Jul 2026 – Present
Simulation-driven state estimation, control & fault-detection portfolio | Python, C++17, MATLAB/Simulink

- Designing and building, in phases, an open-source algorithm portfolio (EKF/UKF estimation, SO(3) geometric control, TCN fault detection) with a CI-tested architecture and requirements/FMEA-FTA baseline, following an ASPICE-style process; implementation ongoing.

Multi-Layer Satellite Constellation Beam & Link Scheduling Optimization Aug 2025 – Dec 2025
Sponsored requirement, China Academy of Space Technology (CAST) | MATLAB

- Modeled satellite selection for a BeiDou-like constellation (24 MEO + 3 IGSO + 5 GEO) as a real-time, event-driven re-selection problem for navigation (4-satellite service, PDOP minimization) and telemetry (single-/dual-hop relay link, delay minimization); implemented a greedy MATLAB algorithm that holds the current assignment while feasible and re-searches all currently visible satellite combinations on failure, with a full LOS/elevation/beam-scan visibility engine and 3D simulation across 30+ heterogeneous users.

Graduate Thesis: VDMS based on MBSE/ARCADIA Methodology Nov 2024 – Mar 2026
SJTU School of Aeronautics and Astronautics | SysML, Capella/ARCADIA

- Built the complete Operational/System/Logical architecture (OA/SA/LA) of a Vehicle Data Management System (VDMS) in SysML using the **ARCADIA** methodology, driven by a 6-stage airworthiness data-lifecycle model addressing ICAO/CAAC data-governance requirements under an IMA architecture.
- Decomposed requirements into **25 atomic functions** and **19 system-level data flows**; designed a 20-component logical architecture (Configuration Policy Manager, Storage Scheduling Gateway, Gateway Sync Service, Analysis Task Manager) with end-to-end data traceability and dynamic policy injection.

Re-entry Spacecraft Space-Segment Simulation & Trajectory Optimization Mar 2024 – Aug 2025
Sponsored requirement, Shanghai Academy of Spaceflight Technology (SAST) | MATLAB, FALCON

- Simulated the space-segment flight dynamics of a re-entry spacecraft from a specified initial orbit to the atmospheric entry interface, and formulated/solved the descent trajectory as a constrained optimal-control problem in MATLAB using **FALCON**, a sequential convex optimization (SCP)-based trajectory-optimization tool.

Technical Skills

Programming: Python, C++
Model-Based Dev.: MATLAB/Simulink, SysML, ARCADIA/Capella, Innoslate
Automotive / Systems: Reliability Workbench (RWB, FMEA/FMECA/FTA), DOORS
Other Tools: PyTorch, Office, LaTeX

Certificates & Languages

- English: CET-6; IELTS exam scheduled for Aug 5, 2026 (current mock score ≈ 7.0). Russian: basic conversational (from the MAI exchange program).